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RESEARCH

Correlation ECE Measurement System for Electron Temperature Fluctuations in LHD

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Abstract

Understanding turbulence behavior in high-temperature plasmas is crucial for elucidating the mechanisms behind anomalous transport phenomena. The correlation electron cyclotron emission (cECE) technique is widely used to measure electron temperature fluctuations associated with turbulence in high-temperature plasmas. A new radial cECE system has been installed in the Large Helical Device (LHD), incorporating tunable bandpass filters (BPFs). Effective spectral decorrelation was obtained by increasing the center frequency separation between two BPFs. As an initial result using the cECE system, relatively low-coherence electron temperature fluctuations were observed in the LHD, with an estimated fluctuation level below 1%.

Keywords Correlation ECE · Temperature fluctuations · High-temperature plasma

Introduction

Elucidating the influence of turbulence on particle and energy transport in fusion plasmas is essential for developing magnetically confined fusion reactors. Ion-scale and electron-scale drift wave turbulence are believed to play a significant role in governing these transport processes [1].

Electron temperature fluctuations $\tilde{T}_e$ caused by the turbulence are a key indicator because they enhance the electron energy flux Q_e^{ES} driven by the electrostatic fluctuations via an interaction with the poloidal electric field $\tilde{E}_\theta$ according to the following equation [2–4]:

$$Q_e^{ES} = \frac{3}{2B_\phi}\left(\bar{n_e}\langle\tilde{T}_e\tilde{E}_\theta\rangle + \bar{T}_e\langle\tilde{n_e}\tilde{E}_\theta\rangle\right), \quad (1)$$

where $\bar{n_e}$ and $\bar{T}_e$ are the ambient electron density and electron temperature, respectively, B_ϕ is the toroidal magnetic field, and $\tilde{n_e}$ represents the electron density fluctuations. Angle brackets denote ensemble averages. Therefore, it is crucial to measure $\tilde{T}_e$ and clarify its characteristics in plasmas to understand the anomalous heat transport mechanism.

The correlation electron cyclotron emission (cECE) method [5] is essential for measuring small $\tilde{T}_e$ in high-temperature plasmas. It has been widely used in various magnetically confined plasma devices, including W7-AS [6], DIII-D [7, 8], Alcator C-mod [9, 10], EAST [11], TCV [12, 13], and ASDEX Upgrade [14]. Several attempts to perform cECE measurements have also been made in the Large Helical Device (LHD) [15] using both the conventional cECE method [16] and the digital cECE method [17–19]. In these studies, the observations of low-frequency $\tilde{T}_e$ below 10 kHz potentially related to turbulence and $\tilde{T}_e$ in the order of tens of kilohertz associated with MHD

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activity were reported. However, high-frequency $\tilde{T}_e$ over 10 kHz induced by turbulence has not yet been conclusively observed, probably because dedicated diagnostic systems were not employed and/or data handling was not sufficiently optimized in previous studies. Therefore, it is important to develop a new cECE system for the LHD to measure high-frequency electron temperature fluctuations caused by drift-wave turbulence.

Section 2 describes the analysis method applied to the data obtained using the cECE system. Section 3 presents the details of the new cECE system developed for the LHD to measure small $\tilde{T}_e$ caused by turbulence. Section 4 shows the initial results obtained with the system. Finally, conclusions are presented in Section 5.

Principle of cECE Measurement

Several correlation radiometry techniques have been proposed to reduce thermal noise and detect the small turbulent fluctuations [5]. The cECE method, which correlates two ECE signals separated in frequency, commonly referred to as spectral decorrelation or radial cECE, has been adopted in numerous devices. The sensitivity limit of the detectable electron temperature fluctuation level for a normal ECE radiometry [6, 20] is determined by

$$\left(\frac{\tilde{T}_e}{\bar{T}_e}\right)_{min} = \sqrt{\frac{2B_v}{B_{IF}}}, \tag{2}$$

where B_{IF} denotes an intermediate frequency (IF) bandwidth and B_v represents a video bandwidth. The detectable limit is approximately 4.5% in the case of $B_{IF} = 1$ GHz and $B_v = 1$ MHz. This sensitivity limit can be improved by using the cECE method and is expressed as [21]

$$\left(\frac{\tilde{T}_e}{\bar{T}_e}\right)_{min} = \sqrt{\frac{2}{\sqrt{N}}\frac{B_v}{B_{IF}}\sqrt{\frac{f_s}{2B_v}}} = \sqrt{\frac{1}{B_{IF}}\sqrt{\frac{2B_v}{\Delta t}}}. \tag{3}$$

Here, N is the number of data points used for the correlation calculation, given by $N = f_s \Delta t$, where f_s is the sampling frequency and Δt is the time duration. The video bandwidth equals the Nyquist frequency $f_s/2$ if the signal is not subject to any frequency limitation, either by hardware or digital signal processing. In this case, Eq. 3 is given by [21]

$$\left(\frac{\tilde{T}_e}{\bar{T}_e}\right)_{min} = \sqrt{\frac{2B_v}{\sqrt{N}B_{IF}}} = \sqrt{\frac{1}{B_{IF}}\sqrt{\frac{f_s}{\Delta t}}}. \tag{4}$$

Therefore, the detectable lower limit using the cECE method is approximately 0.5% in the case of $B_{IF} = 200$ MHz, $f_s = 2$ MHz and $\Delta t = 100$ ms. This is nearly an order of magnitude better than that of the normal ECE radiometry.

Several methods have been proposed to evaluate the electron temperature fluctuation level, such as using cross-correlation [6], cross-power spectral density (CSD) [22], and coherence [14]. A recent study [23] showed that the power spectral density (PSD) of the $\tilde{T}_e$ level obtained using the radiometers with different IF bandwidths, B_{IF1} and B_{IF2}, can be expressed as

$$P_{\tilde{T}_e}(f) = \frac{\gamma^2\left(\frac{1}{B_{IF1}} + \frac{1}{B_{IF2}}\right) + \sqrt{\gamma^4\left(\frac{1}{B_{IF1}} - \frac{1}{B_{IF2}}\right)^2 + \frac{4\gamma^2}{B_{IF1}B_{IF2}}}}{2\left(1-\gamma^2\right)}. \tag{5}$$

Here, γ is the coherence between two ECE signals and is defined as [24]

$$\gamma = |\gamma_{xy}| = \frac{|P_{xy}|}{\sqrt{P_x P_y}}, \tag{6}$$

where γ_{xy} is the complex coherence, P_{xy} is the CSD, and P_x and P_y are the auto-PSDs of the two signals. The significance level of the coherence [25] is defined as

$$\gamma_{SL} = \sqrt{1 - \alpha^{1/(n_d - 1)}}, \tag{7}$$

where α is the significance level (Type I error), representing the probability of incorrectly rejecting the null hypothesis. A value of $\alpha = 0.05$ (5%) is conventionally adopted. Here, n_d is the number of segments to calculate the CSD and PSD without overlapping. If the condition of $B_{IF} = B_{IF1} = B_{IF2}$ is satisfied, the expression of $P_{\tilde{T}_e}$ is simplified as [23]

$$P_{\tilde{T}_e}(f) = \frac{\gamma}{B_{IF}\left(1-\gamma\right)}. \tag{8}$$

The coherence can be modified to derive a more accurate result by subtracting a bias error of coherence γ_b [24] as follows [23]:

$$\gamma_c = \sqrt{\gamma^2 - \gamma_b^2}, \tag{9}$$

$$\gamma_b^2 \approx \frac{\left(1-\gamma^2\right)^2}{n_d}. \tag{10}$$

In our analysis, the bias-corrected coherence γ_c is set to zero when γ_b exceeds γ. The $\tilde{T}_e$ level induced by turbulence can be estimated by integrating its PSD over the frequency range from f_1 to f_2, where the turbulence components are present, as follows:

$$\frac{\tilde{T}_e}{\bar{T}_e} = \sqrt{2\int_{f_1}^{f_2} P_{\tilde{T}_e}(f)df}. \tag{11}$$

Welch's method [26] was adopted to calculate both auto-PSD and CSD, employing a Hann window with 50% overlap between adjacent segments to improve the accuracy of the coherence estimation because the finite segment overlap reduces the bias error and the variance [27]. The equivalent degree of freedom (EDOF) ν using the Hann window with 50% overlap is expressed as follows [28]:

$$\nu \approx \frac{2m_d}{1+2(1-1/m_d)\left|\Sigma_{t=1}^{n_s/2} h_t h_{t+n_s/2}\right|^2} \approx \frac{36m_d^2}{19m_d-1} \tag{12}$$

where n_s is the number of data points per segment, m_d is the number of segments accounting for the overlap, and h_t is the normalized Hann window. The equivalent number of averages (ENA) is half of the EDOF because two degrees of freedom are contributed by each segment [24]. In this analysis, the ENA is used in place of n_d in Eqs. 7 and 10. Furthermore, the $\tilde{T}_e$ level is estimated using the PSD formulation consistent with Eq. 5.

Measurement Setup of cECE in LHD

A new cECE system was developed and installed in the LHD. Figure 1 illustrates the configuration of the cECE circuit. The optical system which is shared with Q- and V-band ECE radiometers [29], consists of a focusing mirror and a flat mirror inside the vacuum vessel to transmit

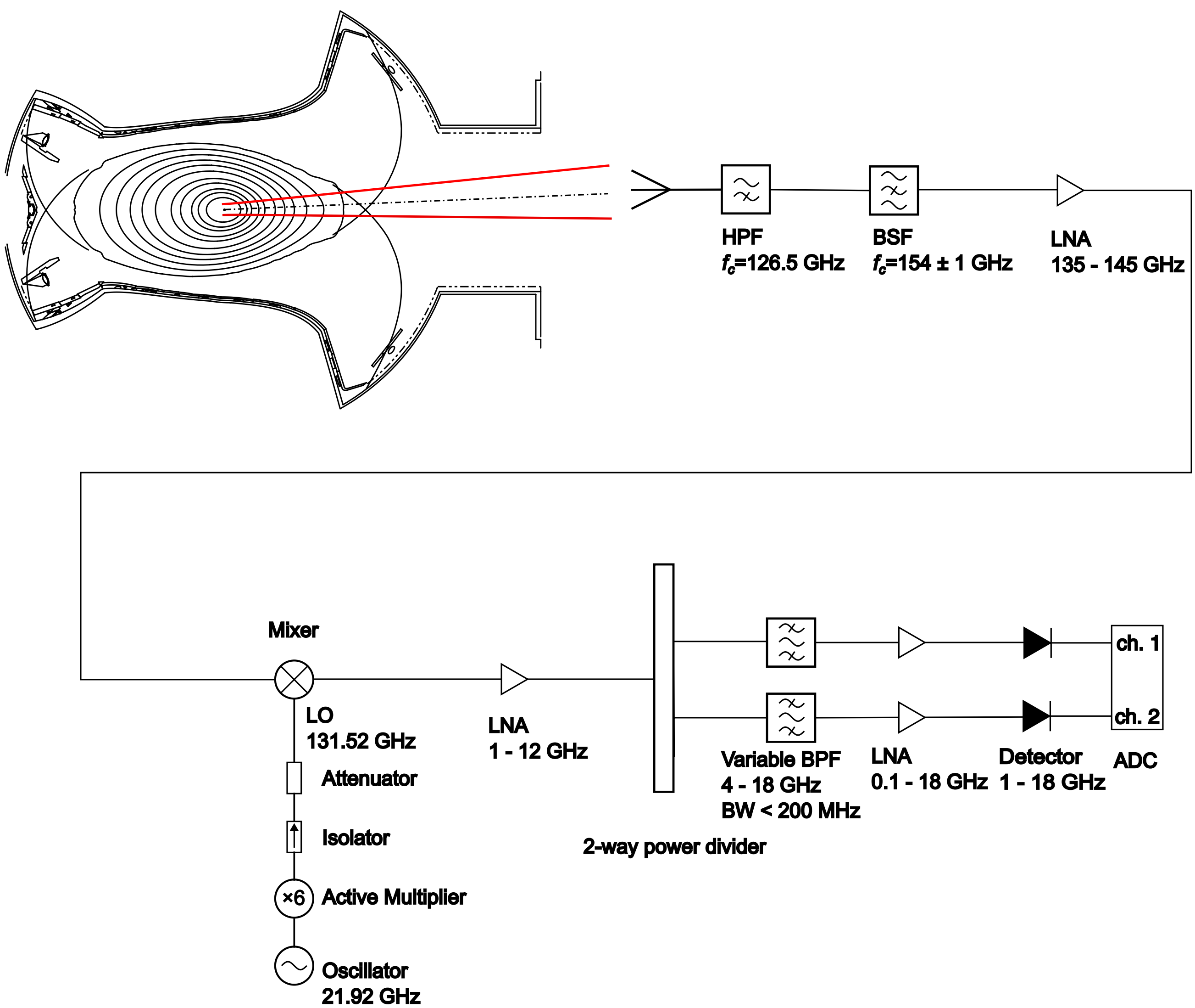

Fig. 1 Configuration of the cECE circuit in the LHD

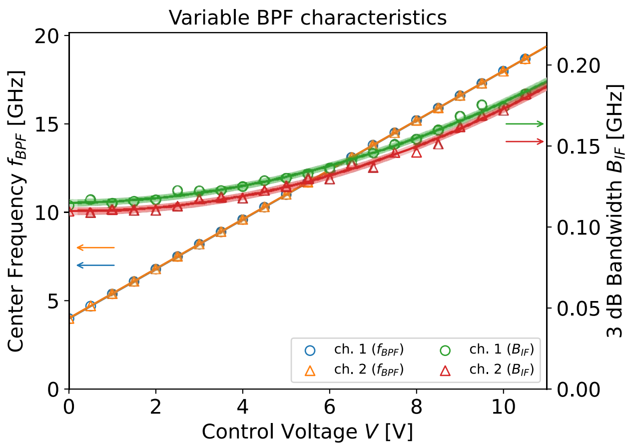

Fig. 2 Dependence of f_{BPF} and 3 dB bandwidth B_{IF} of the YIG BPF on control voltage. Blue and green circles indicate the measured f_{BPF} and B_{IF} for ch. 1, respectively, and yellow and red triangles indicate those for ch. 2, respectively. Solid lines represent regression curves fitted to each data set, and the shaded areas show the ranges within one standard deviation

the signal to a waveguide through a vacuum window, and the sightline passes through the horizontally elongated cross-section of the LHD. The received ECE microwaves are transmitted to a mixer via a high-pass filter (HPF), a bandstop filter (BSF), and a radio-frequency (RF) low-noise amplifier (LNA). The BSF, developed at the National Institute for Fusion Science, attenuates strong leakage power from the 154 GHz gyrotrons and protects the downstream components. Its stop-band is centered at 153.9 GHz with a bandwidth of 1.9 GHz. In the RF section, signals from 135 GHz to 145 GHz are amplified by the RF LNA and then down-converted to IF signals by the mixer, using a local oscillator (LO) set at 131.52 GHz. In the IF section, the signals are further amplified, split into two channels, filtered by a variable yttrium iron garnet (YIG) bandpass filter (BPF), re-amplified, detected, and recorded by a digitizer. Figure 2 shows the dependence of the BPF center frequency f_{BPF} and B_{IF} on the control voltage for channels 1 (ch. 1) and 2 (ch. 2). B_{IF} increases moderately with f_{BPF} and exhibits a slight difference between the channels. The difference of B_{IF} can be taken into account when estimating the PSD of the $\tilde{T}_e$ level using Eq.5. The system covers ECE signals in the frequency range from 135 GHz to 143.52 GHz under the present circuit configuration. The resolution of the poloidal wavenumber k_θ is limited by the antenna optics, whereas that of the radial wavenumber k_r is determined by the radial size of the emission volume [5]. At 135 GHz, a Gaussian beam with a waist $w \approx 13$ mm (the maximum value within the observed frequency range) is received by the optics at the radial position $R \approx 4.1$ m and the vertical position $Z \approx 0$ m. The poloidal wavenumber resolution is calculated as $\Delta k_\theta = 2\pi/2w \approx 0.25\ \mathrm{mm}^{-1}$. Using the ray-tracing code TRAVIS [30], the radial spatial resolution ΔR is estimated to be approximately 17 mm at 143.5 GHz for the standard LHD magnetic configuration of the magnetic axis $R_{ax} = 3.6$ m and the magnetic field $B = 2.75$ T, assuming $n_e \approx 1.5 \times 10^{19}\ \mathrm{m}^{-3}$ and $T_e \approx 1.5$ keV near the emission region. Consequently, the radial wavenumber resolution is limited to $\Delta k_r = 2\pi/\Delta R \approx 0.37\ \mathrm{mm}^{-1}$ for this specific parameter set.

The performance of the BPFs was evaluated using a noise source to validate that thermal noise can be sufficiently reduced by increasing the center frequency separation. Figure 3(a) shows the relationship between the cross-correlation coefficient function $C_{xy}(\tau)$ at lag time $\tau = 0$ and the center frequency difference of BPFs, Δf_{BPF}, around $f_{BPF} = 12$ GHz. The value of $C_{xy}(0)$ decreased from nearly 0.8 as Δf_{BPF} increased, and converged to

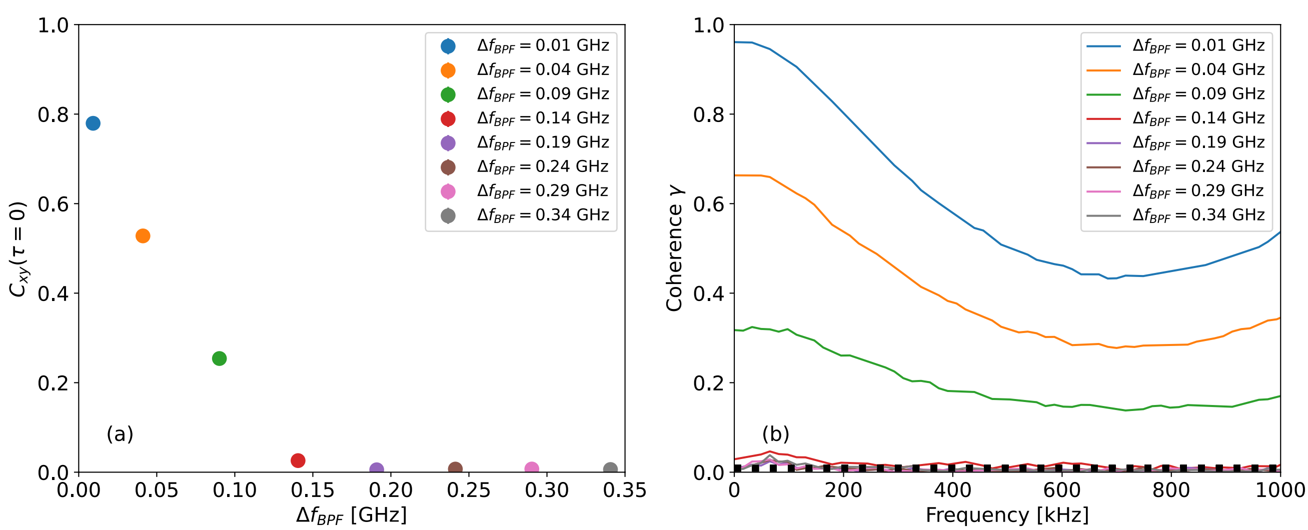

Fig. 3 (**a**) Relationship between $C_{xy}(\tau = 0)$ and Δf_{BPF} and (**b**) comparison of coherence spectra for various Δf_{BPF} with f_{BPF} of ch. 1 fixed at approximately 12 GHz. The dotted line in (**b**) indicates γ_{SL} corresponding to the 5% significance threshold

zero at $\Delta f_{BPF} > 150$ MHz. This value is comparable to B_{IF} near 12 GHz. Figure 3(b) shows the coherence of each Δf_{BPF}. The coherence decreased with the increase of Δf_{BPF} and approached γ_{SL} for $\alpha = 0.05$ in the range of $\Delta f_{BPF} > 150$ MHz. These results validate that sufficient thermal noise reduction can be achieved by adjusting the center frequency separation of the BPFs.

Experimental Results

The measurement of $\tilde{T}_e$ was conducted using the new cECE system in the LHD plasma of $R_{ax} = 3.6$ m and $B = 2.75$ T. Figure 4 shows the time evolution of ECH and NBI power, ECE signals measured by the cECE system, and the electron temperature (T_e) and density (n_e) measured by the Thomson scattering system [31]. The center frequencies of the BPFs were set at 11.99 GHz for ch. 1 and 11.7 GHz for ch. 2 and the received signals corresponded to approximately 143.5 GHz and 143.2 GHz, repsectively. During this discharge, T_e decreased as the ECH power was reduced. As expected, the raw signals measured by the cECE system exhibited a time evolution similar to that of T_e, despite the different measurement locations.

Two time intervals were selected to investigate the PSD of the $\tilde{T}_e$ level: an ECH-sustained plasma period from 3.8 s to 6.3 s (ECH phase) and a primarily NBI-sustained plasma period from 8.2 to 9.8 s (NBI phase). These intervals are indicated by red and blue hatched regions in Fig. 4, respectively. Figure 5 shows the radial profiles of n_e and T_e at 5.0003 s and 9.0002 s, corresponding to representative time slices from each phase. In Fig. 5, the horizontal axis represents the normalized minor radius $\rho = r_{eff}/a_{99}$, where r_{eff} is the effective minor radius and a_{99} is the averaged minor radius enclosing 99% of the plasma stored energy. Using the TRAVIS code, the radial positions for the ECE frequencies of 143.5 GHz and 143.2 GHz were estimated as $\rho \approx 0.34$ and 0.35 during the ECH phase (at 5.0003 s), and $\rho \approx 0.26$ and 0.27 during the NBI phase (at 9.002 s), respectively. In both phases, the plasma was optically thick, with the optical depths for both frequencies exceeding 10, the value of ΔR ranged from 14 mm to 17 mm, and the radial separation between the two measurement locations was less than 8 mm. This spatial separation is comparable to the typical scale of density fluctuations with wavenumbers ranging from 0.1 to 1.0 mm^{-1}, as measured by 2D-PCI in the LHD [32].

Figure 6 shows the CSD, phase difference of the ECE signals, coherence, and PSD of the $\tilde{T}_e$ level during the ECH and NBI phases. Each segment used for the spectral analysis contained 512 data points. The modified coherence and the PSD of the $\tilde{T}_e$ level estimated using the modified coherence are also shown. The spectral analysis results reveal distinct characteristics between the ECH and NBI phases. The CSDs during both phases were higher than the one during the no-plasma period (0–2 s) for frequencies below 200 kHz. The phase difference remained relatively constant up to ∼50 kHz during the ECH phase, whereas it exhibited significant variations during the NBI phase. The coherence and the PSD of the $\tilde{T}_e$ level in the low-frequency range were higher

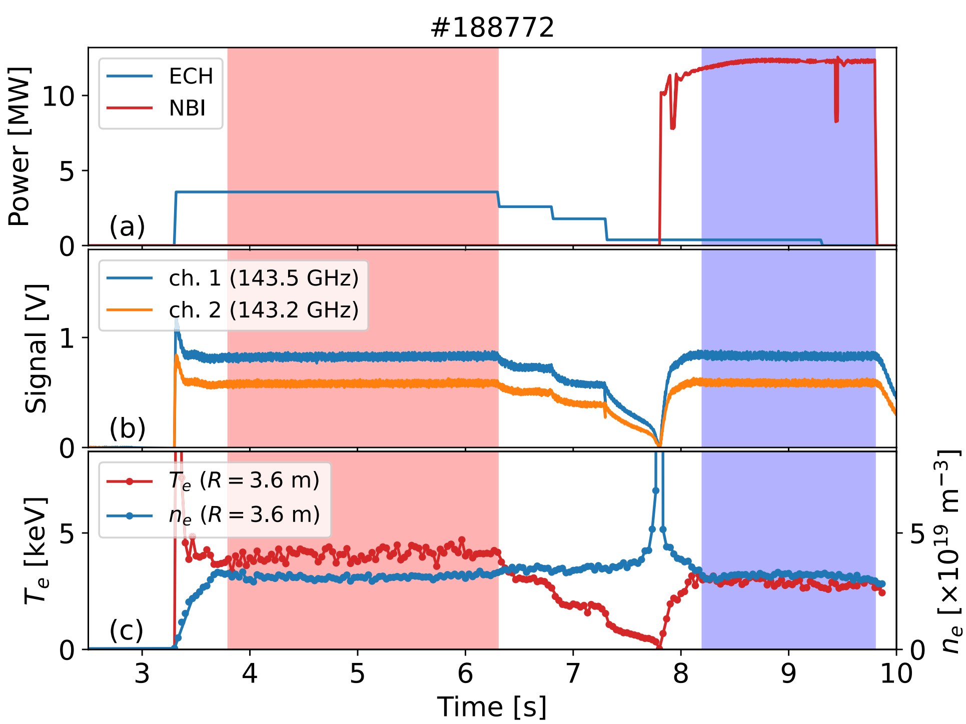

Fig. 4 Time evolution of (**a**) (blue) ECH and (red) NBI power, (**b**) raw ECE signals measured by (blue) ch. 1 and (orange) ch. 2 of cECE system, and (**c**) (red) T_e and (blue) n_e measured by Thomson scattering around $R = 3.6$ m. The red and blue hatched regions indicate the time duration when the data were analyzed in Fig. 6

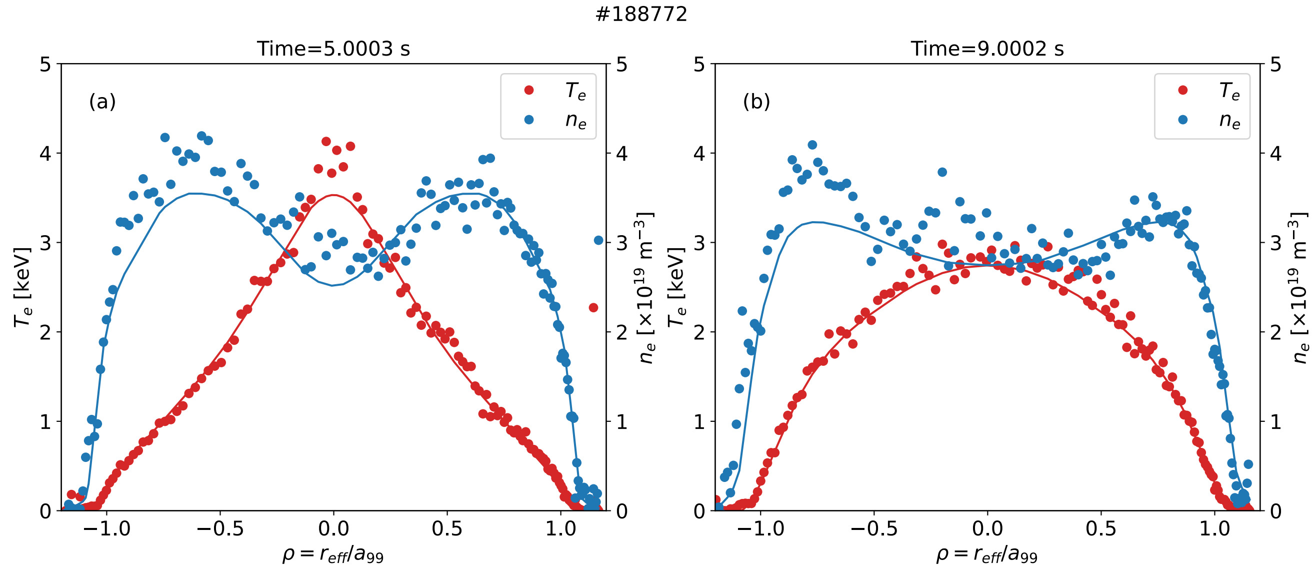

Fig. 5 Radial profiles of (red) T_e and (blue) n_e measured by Thomson scattering at (**a**) 5.0003 s (ECH phase) and (**b**) 9.0002 s (NBI phase)

during the ECH phase than during the NBI phase. In both phases, the observed coherence values remained below 0.1. During the ECH phase, only the relatively low-frequency components of the coherence exceeded γ_{SL}, which was the 5% significance level, while nearly all coherence values below 200 kHz during the NBI phase were lower than γ_{SL}.

Figure 7 shows the $\tilde{T}_e$ levels estimated from γ and γ_c during the ECH and NBI phases. Each fluctuation level was calculated using Eq. 11 with $f_1 \approx 9.8$ kHz and $f_2 \approx 53.7$ kHz. During the ECH phase, the $\tilde{T}_e$ level estimated from γ was approximately 0.34%, while that estimated from γ_c was naturally lower, at approximately 0.32%. During the NBI phase, the $\tilde{T}_e$ levels estimated from γ and γ_c were approximately 0.26% and 0.20%, respectively. In all cases, the estimated values were above the sensitivity limit (approximately 0.12% during the ECH phase and 0.14% during the NBI phase), as determined from Eq. 3, where $B_{IF} \approx 128$ MHz at 143.2 GHz and the video bandwidth is given by $B_v = f_2 - f_1 \approx 43.9$ kHz [21]. In contrast to the ECH phase, the difference between the $\tilde{T}_e$ levels estimated from γ and γ_c was more pronounced during the NBI phase because the lower coherence during this phase made the effect of bias error subtraction relatively significant. Applying the coherence bias subtraction helped to clarify the difference in the estimated $\tilde{T}_e$ levels. The milder n_e and T_e gradients around the magnetic axis in the NBI phase may suppress the $\tilde{T}_e$ level relative to the ECH phase. By using γ_{SL} at $\alpha = 5\%$ as the threshold value of coherence for the $\tilde{T}_e$ level estimation, almost all influence of the zero-coherent signals can be rejected, and overestimation can thereby be avoided. However, the real coherence values lower than γ_{SL} can also be dismissed. Provided that the $\tilde{T}_e$ level was estimated by integrating the PSD only over the frequency range where $\gamma > \gamma_{SL}$, the $\tilde{T}_e$ levels estimated using γ and γ_c were approximately 0.31% and 0.30%, respectively, during the ECH phase, while those during the NBI phase were 0.19% and 0.18%. If a conservative estimate of the $\tilde{T}_e$ level is required, using γ_{SL} as the threshold may be helpful.

Conclusion

The new cECE system utilizing variable BPFs was installed in the LHD to measure small $\tilde{T}_e$ induced by turbulence. This system validated that thermal noise can be sufficiently reduced by adjusting Δf_{BPF}. By using the cECE system, initial experimental results were obtained, and the PSD of the $\tilde{T}_e$ level was derived from the measured coherence, which showed clear differences between the ECH and NBI phases. The observed fluctuations exhibited relatively low coherence (below 0.1), and the estimated $\tilde{T}_e$ level was less than 1%.

Expanding the RF frequency range will allow access to the plasma edge region, where clearer and more coherent signals may be observed because the larger $\tilde{T}_e$ level is expected. Furthermore, if the radial profile of the $\tilde{T}_e$ level from the edge to the core can be measured, it will significantly contribute to advancing energy transport studies in the LHD.

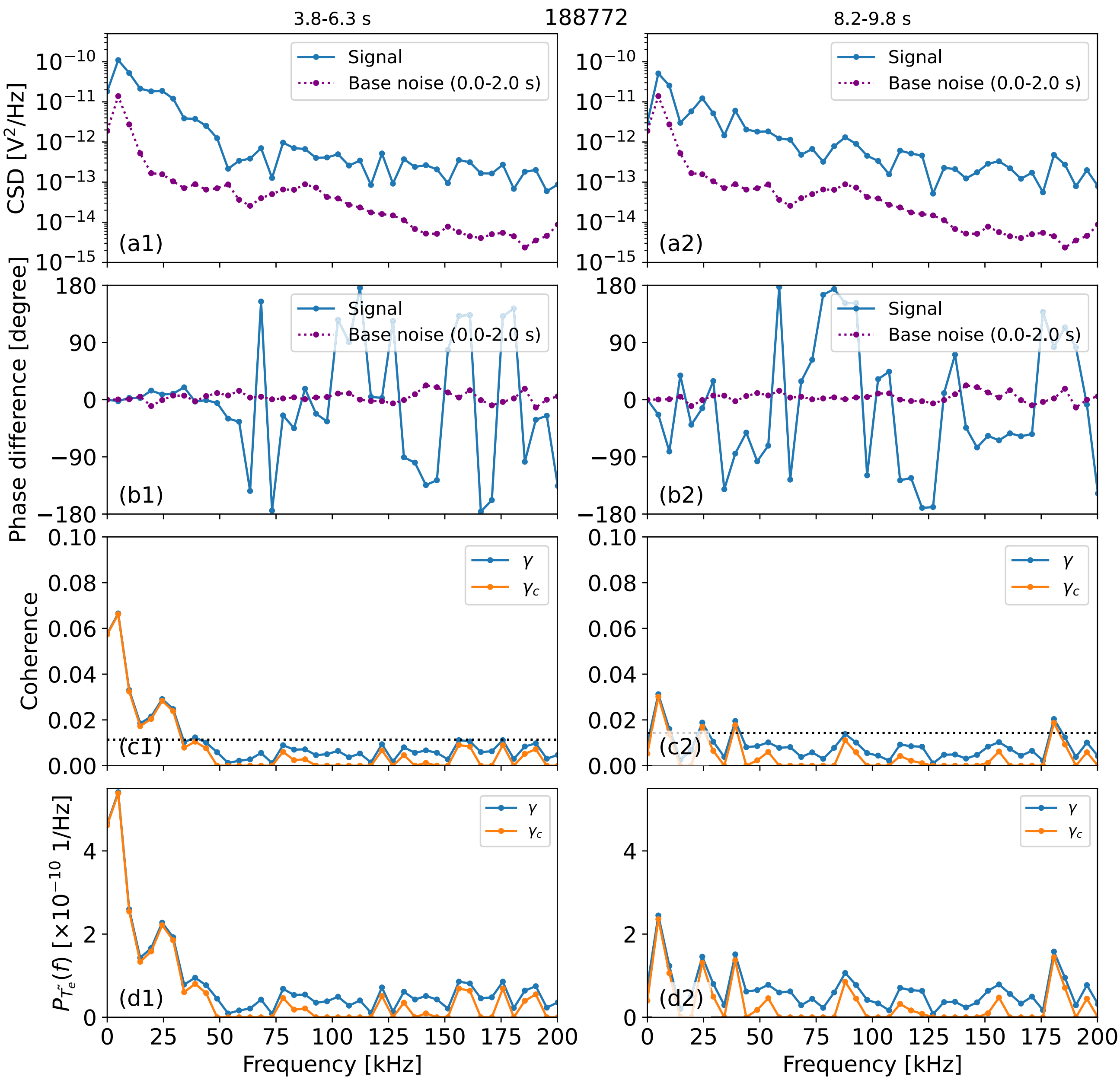

Fig. 6 (a1, a2) CSDs, (b1, b2) phase differences, (c1, c2) coherences and modified coherences, (d1, d2) PSDs of the $\tilde{T}_e$ levels. Blue lines indicate the values during the plasma periods of the ECH phase (3.8–6.3 s) and the NBI phase (8.2–9.8 s). Purple lines represent the CSD and phase difference during the non-plasma period (0–2 s). Orange lines represent γ_c and the corresponding PSDs. The black dotted lines in (c1) and (c2) represent γ_{SL}, which corresponds to the 5% significance threshold

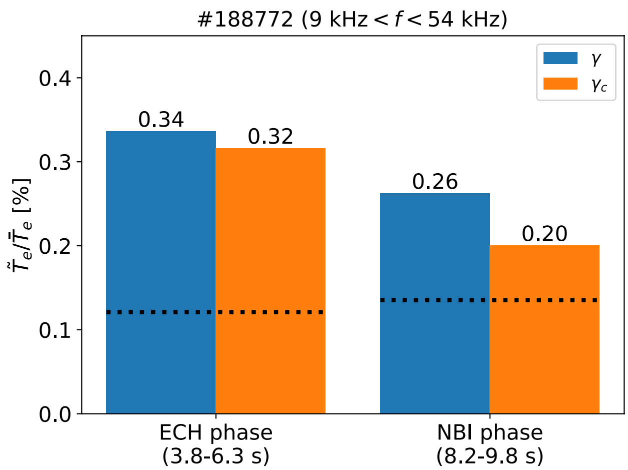

Fig. 7 $\tilde{T}_e$ levels during the ECH phase (3.8–6.3 s) and the NBI phase (8.2–9.8 s), calculated from the PSDs of the $\tilde{T}_e$ level using (blue bars) γ and (orange bars) γ_c. Each level was obtained by integrating the PSD over the frequency range of approximately 9–54 kHz. The black dotted lines indicate the sensitivity limits

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Author Contributions R. Y. wrote the main manuscript text and analysed experimental data. T. T., T. N., M. N., and D. N. contributed to the preparation of the diagnostic system. All authors reviewed the manuscript.

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Declarations

Competing interests The authors declare no competing interests.

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